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Creating Volcano Plots From Loupe Genomic Data

Computational biology is a rapidly developing field that can greatly aid many aspects of biology, including ecology, physiology, and microbiology, but for the primary purpose of my project, genetic health biology. In computational biology, there are many advanced and complex methods for visualizing data, especially genomic data. A program that does this very well is the 10x Genomics Loupe Browser. This program takes genomic data from a tissue sample and clusters cells based on gene expression differences. These clusters can be shown in various ways: K-means, t-SNE, and spatial. These methods are all similar, but some highlight different desired features of the data. While this visualization is very powerful, it can be a lot to take in, as it presents a lot of data at once, and it may be hard to investigate specific genes that lead to cluster formation. In my project, I created a Jupyter notebook where you can download 10x Genomics Loupe data and use volcano plots to identify more specific genes driving cluster differentiation. To better illustrate how my notebook works, I will use a breast cancer sample to demonstrate how the technology can be used.

Cancer is the second leading cause of death in the United States, causing around 1 in 6 deaths, equating to roughly 9.7 million deaths a year (1). There are two common causes of cancer: the first is oncogenes, which are mutated genes and are now overly active, producing tumor tissue. Secondly, there are mutated tumor suppressor genes, such as p53. When the gene is mutated, it becomes inactive and unable to suppress other genes that promote cancer cell growth. Cancer has varying levels of severity, often determined by how “differentiated” the cancer is.

Undifferentiated cells are those that have lost their specialized, mature structures and functions and often exhibit very random gene expression. These tumors that are highly undifferentiated are much more aggressive and are harder to treat, often becoming a lot more deadly. (2) Looking at the breast cancer sample from the 10x Genomics database, there are many different clusters of cells scattered throughout the spatial visualization of the tissue (Figure 1). While we can clearly see the different clusters of the sample and their gene expressions in the browser, there is a lot to look at all at once, and we cannot compare which genes truly differentiate the clusters. That being said, we can still make a hypothesis about this sample using the spatially visualized data. For example, since there are many different oddly shaped clusters all mixed in with each other, I believe this is a highly undifferentiated tumor sample, leading to more difficult treatment of the patient. To investigate whether or not this sample is highly undifferentiated, we can use my Jupyter notebook to investigate the genes contributing to the regulation and identity of each cluster.

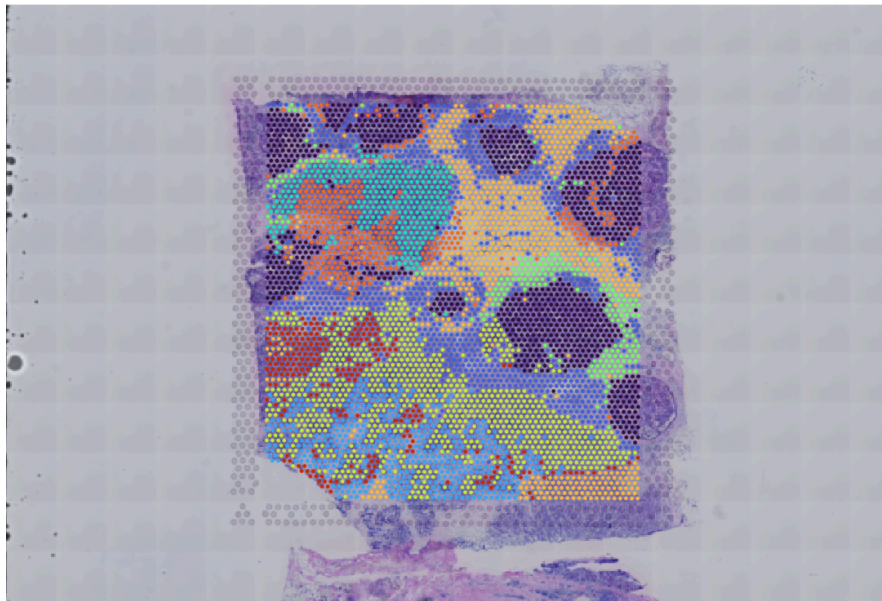


Figure 1. Human breast cancer sample in a spatial view on the 10x Genomics Loupe browser.

The details of how I created the Jupyter notebook, along with all the instructions, are in the notebook itself, but for this paper I will focus on the results. After generating volcano plots for all the clusters from the sample using my notebook, I chose to highlight Cluster 7. In the volcano plot for Cluster 7, we can see many genes that are both upregulated and downregulated relative to the other clusters. The most significantly upregulated genes are AFP, CARTPT, and FSIP1 (Figure 2).

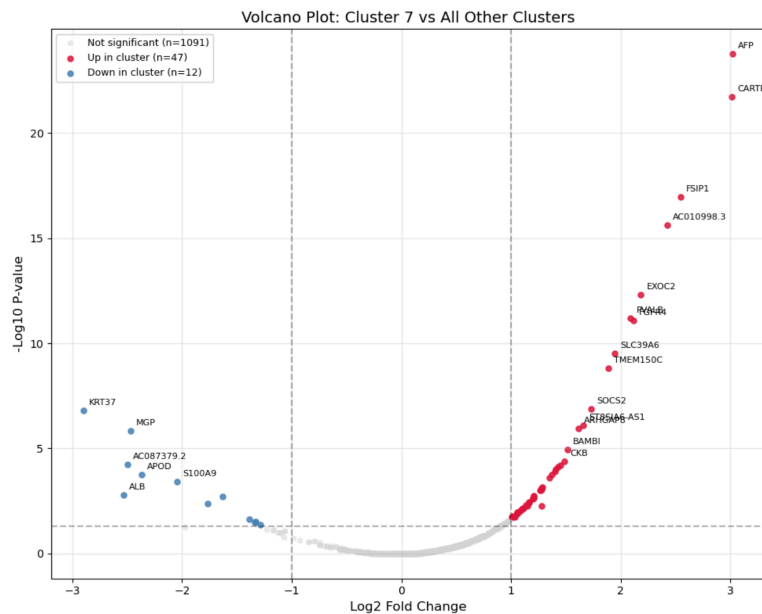


Figure 2. Volcano plot of Cluster 7 gene expression data from the 10x Genomics database generated using my notebook of

In other clusters, there were similar results, but some also showed varying levels of unique up- and down-regulation. Where the results are highly similar across all clusters is in the types of genes expressed. The three in Cluster 7 are all very different and quite random: AFP related to yolk-sac formation, CARTPT related to endocrine function, and FSIP related to sperm mobility. These genes do not relate to each other in terms of function; however, since they are all expressed in the sample, this suggests that the sample is, in fact, a highly undifferentiated cancer

tumor that lacks any original function. Since the majority of the work on this project focused on the notebook itself, and using breast cancer as an example, I did not delve too deeply into the research for every gene expressed. However, all the clusters do seem to support my hypothesis that this is a highly undifferentiated cancer sample, as they all contain genes related to diverse physiological functions. I should note that to ensure the notebook's reliability, I vetted it by cross-referencing the original gene expression data against the original visualization shown in the volcano plots. Confirming that the most significantly up- and downregulated genes (such as AFP, CARTPT, and FSIP1 in Cluster 7) matched the expression patterns in the source data verified that the notebook is accurately processing and visualizing the underlying genomic information.

The results support my hypothesis that the breast cancer sample is a highly undifferentiated tumor. The unrelated functions of the most upregulated genes in Cluster 7, along with similarly disparate expression patterns across the other clusters, suggest a tissue that has lost its specialized identity, consistent with the aggressive, hard-to-treat nature of undifferentiated cancers. Ultimately, this project demonstrates how my Jupyter notebook can extend the capabilities of the 10x Genomics Loupe Browser by allowing researchers to investigate the specific genes causing cluster differentiation through volcano plots. While the Loupe Browser provides a strong overview of spatial and clustered genomic data, the notebook adds a more focused, statistically meaningful layer of analysis, enabling the transition from a general hypothesis to gene-level evidence. Overall, the notebook provides a reproducible, gene-level workflow that complements existing visualization tools and can be applied to any 10x Genomics dataset.

AI Note: I only used AI to research more specific genes and used Grammarly for grammar which I believe uses AI but only punctuation and typo changes.

Citations:

1. American Cancer Society. (n.d.). *Cancer facts & statistics*. Retrieved March 17, 2026, from <https://www.cancer.org/research/cancer-facts-statistics.html>
2. Izar, B., Tirosh, I., & Stover, E. H. (2024). Single-cell sequencing: A guide for the curious clinician. *PMC/National Library of Medicine*. Retrieved May 5, 2026, from <https://pmc.ncbi.nlm.nih.gov/articles/PMC10883260/>